Assignment - 1 DBMS

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Section - A

1- What is DBMS ? Why is it required ?

A Database management system (DBMS) is a software that allow users to store, manage, retrive and manipulate users to store, manage, return, retrive and manipulate data efficiently.
- It is required because it ensures data security, maintains data consistency, Provides easy data access, reduces data redundancy, supports backup & recovery, manages concurrent access.

2- Why is DBMS better than the traditional file system ?

DBMS is better than the traditional file system because it reduces data redundancy. ~~which~~ but in file system data is often duplicated in multiple file which causes inconsistancy. In file system Accessing data requires writting custom programs for each query but DBMS provides query languages which makes easy data Access. DBMS also provides support backups & recovery, security and data integrity.

3- What is data abstraction ? Explain its levels.

Data abstraction is hiding the complete details of data storage and showing only what the user needs.
Levels of Data abstraction are :
1- Physical level : How data is actually stored (disk, files, indexes).
2- Logical level : what data is stored and relationships (tables, fields)
3- View level : what a user sees, only relevant data (custom views).
eg : A teacher sees only student name and mark, not full database details.

4- What is the three - schema architecture ? Why is it important ?

It organises a database into three levels to separate storage, structure, and user views levels :
1. Internal schema (Physical level) : How data is stored (files, indexes).
2. Conceptual schema (logical level) : what data exists and relationships (tables).
3. External schema (View level) : what users see.

It is important because it provides data abstraction, ensures data independence and ~~th~~ enhances security and maintainance.

5- What is data Independence? why is it needed?

Data independence is the ability to change the database structure at one level without affecting other level without affecting user applications.

Types of data independence is logical (changing the logical schema) and physical (changing the physical storage).

It is needed because it reduces maintanance effort, improves flexibility and enhances security and abstraction.

6- What is an Entity - Relationship (ER) Model?

The ER model is a way to visually represent the structure of a database.

It shows:

- Entities (things or objects)
- Attributes (properties of entities)
- Relationships (how entities are connected)

7- What are integrity constraints? Why are they important in DBMS?

Integrity constraints are rules that ensure the correctness, accuracy and consistency of data in a database. They prevent invalid or inconsistent data from being stored.

The types of integrity constraints are Entity, Referential, Domain and user-defined constraints. It is important in DBMS because it maintains accuracy and prevents invalid or inconsistent data and reduce errors.

Section-B Design, Draw & Defend

8. Draw the three schema architecture and explain each level.

External level
(what users see)
(View schema) ← User views

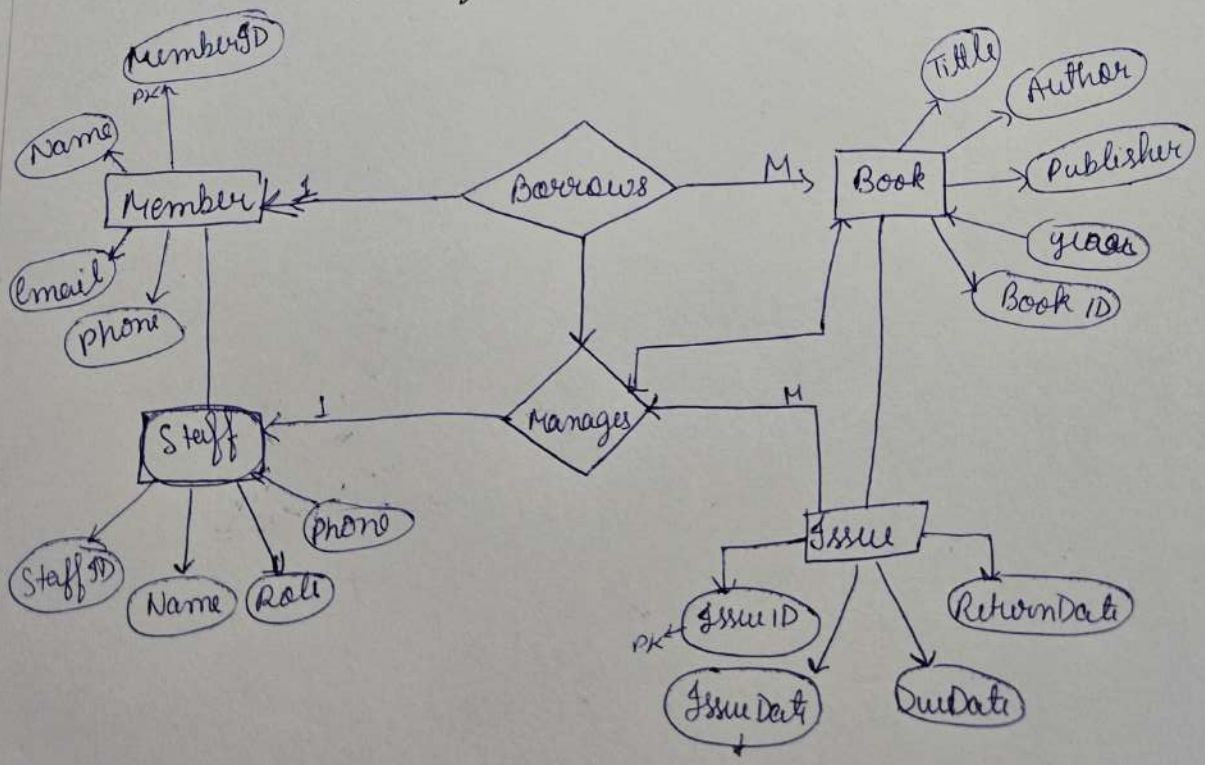
↑
↓ Mappings

Conceptual level (show what data exists)
(Database Structure and relationships) ← Logical schema

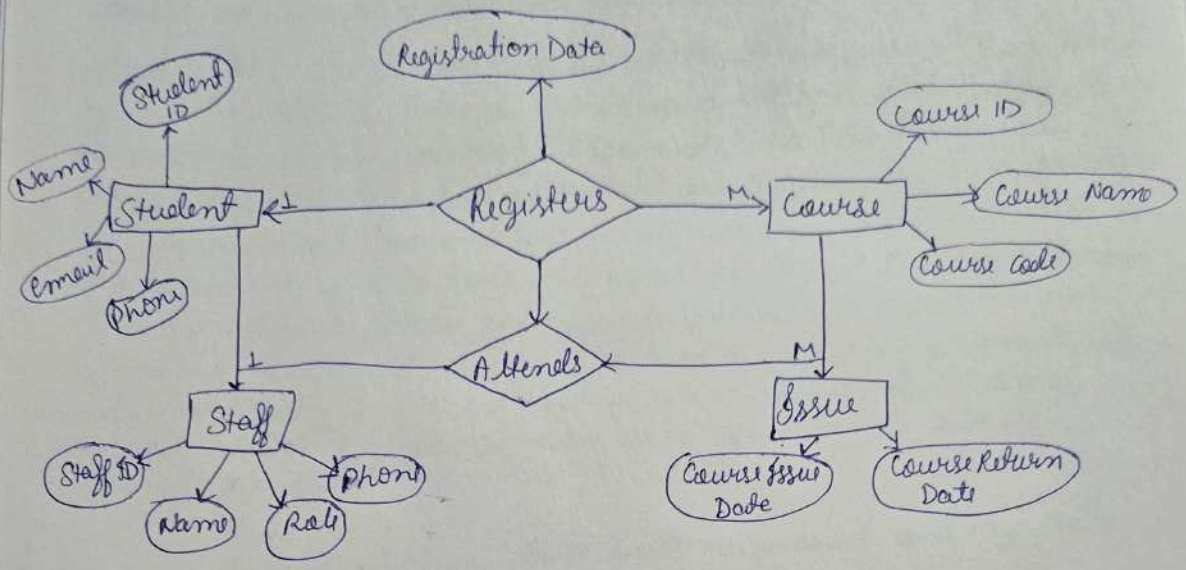
↑
↓ mappings

Internal level (Describe how data is physically stored)
(Physical storage, files, indexes) ← Physical schema

9. Draw an ER diagram for a library Management system.



10. Draw an ER Diagram for a student course Registration system.



Section-C case based analytical questions

11. A hospital wants to store patient, doctor, and appointment data. Identify entities, attributes and relationships.

In this entities are patient and its attributes are patient ID, Name, Age, Gender, Address, Phone and A Doctor is entity and its attributes are DoctorID, Name, speciality, Phone and Email and one more entity appointment and its attributes are appointment ID, appointment Date, appointment Time and here foreign keys are PatientID which link to patient and DoctorID which links to Doctor. And here relation-ships are between patients and appointment because A patient can have many appointment because A patient can have many appointments and its cardinality is 1:M. And another relationship is between doctor and appointment. A doctor can have many appointments and its cardinality is 1:M.

12. An online shopping platform stores customers product, and key order data. Suggest keys and constraints.

Entities & keys:

1. Customer - CustomerID (PK), Name, Email (Unique), Phone (Unique), Address.
2. Product - ProductID (PK), Name, Price
3. Order - OrderID (PK), OrderDate, CustomerID (FK)
4. OrderItem - OrderItemID (PK), OrderID (FK → order), ProductID (FK), Quantity, Price.

Relationship are between customer and order which is 1:M and order and Product which is M:M.

Here constraints are Primary key must be unique for each entity. And foreign key maintains relationships and email and phone must be unique and check price should be greater than zero.

13. A university stores student records. Explain how data abstraction helps different users.

Data abstraction helps by hiding unnecessary details from users and showing only what they need, making the database easier to use and maintains has three levels:

Physical Level (Lowest level) - Shows how data is stored physically on disk, like files, indexes.
ex: The database stores student records as block on a server

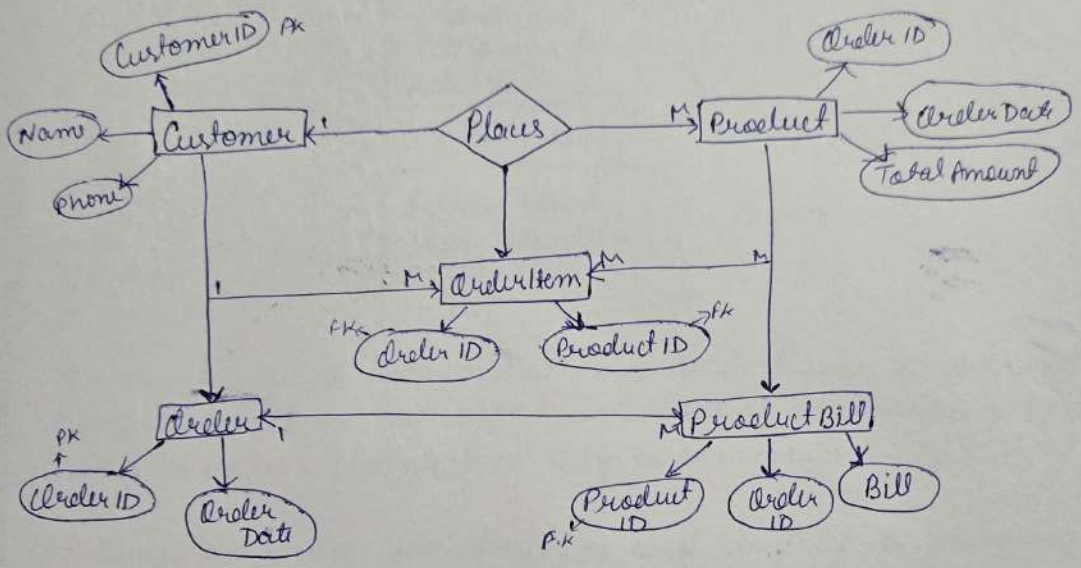
Logical Level (Conceptual Level) - Shows what data exists and how its related without worrying about physical storage.
ex: Tables student (ID, Name, Age, Department) and Course (courseID, Name) and their relationships.

View level (External level) :- Show ~~you~~ only the part of data a user needs.
Different users see different views.
ex: A professor sees student grades in their courses and A student sees only their own grades and courses.

Section – D One diagram + One insight

14. Draw a simple ER diagram for any real-life system and explain one key design decision.

In this my entities are customers, Order and product and relationship places between customer and order. And here key design decision is using an order item table (or M:M relationship) between order & product.

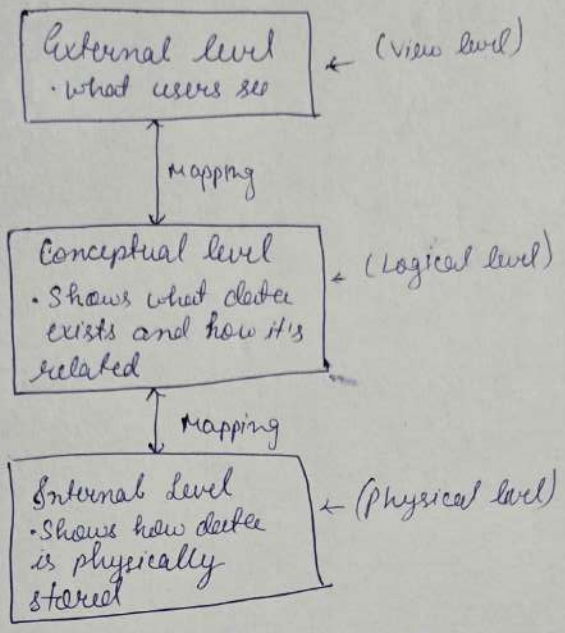


Justification:
An order can have multiple products and a product can appear in many order. Using these steps and this keeps database scalable and easy to maintain.

15. Draw the three schema architecture and mention one benefit.

It is a way to organize a database into three levels to seperate user views, logical structure, and physical storage, which enables data abstraction and database Independence.

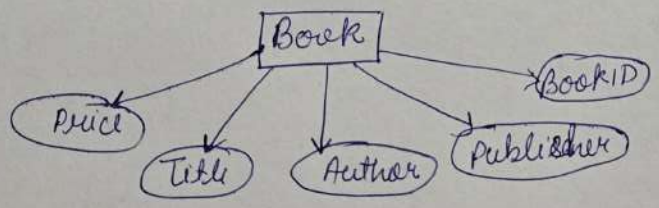
Levels :-



Its key Benefit is Data Independence as it changes at internal or conceptual level do not affect user views or applications which makes it secure, flexible and easy to maintain.

16. Draw an entity with attributes and identify its primary key.

Entity example: Book



In this example, BookId is the primary key because its ensures that every book is unique and other attributes like Title, Author, Publisher provide details but do not uniquely identify a book.